

The Collateral Impact of the Three-Dimensional Jacobian Counterexample: a Source-Verified Cascade

Felipe Santibañez-Leal

CAOS Research program, github.com/fsantibanezleal/CAOS_RESEARCH

2026-07-22 (companion document, v0.02: the explicit dimension-48 witness added)

Abstract

On 2026-07-19, Levent Alpöge announced an explicit counterexample to the Jacobian conjecture in dimension 3, found working with the language model Claude Fable (Anthropic). The credit for every consequence documented here belongs to that single map. This companion records what else that map decides: through implication chains published over five decades, and re-verified against their primary sources by an experiment-history program (records EXP-001, EXP-016, EXP-018), the counterexample resolves or partially resolves the Mathieu conjecture, the Dixmier conjecture, the Poisson conjecture, the Gaussian moments conjecture, Zhao’s vanishing conjecture, the Image conjecture, and the symmetric (gradient) form of the Jacobian conjecture. We state each corollary with its chain and its reference, list precisely what remains open, and describe the verification method, whose value is exactly that a one-line certificate could be turned into a reliable status map of a whole neighborhood of problems within two days. New in this revision: the symmetric/gradient and vanishing rows are upgraded from existential to EXPLICIT: a non-invertible gradient Keller map $\text{id} + \nabla f$ on $\mathbb{C}^{48}$, with f a homogeneous Hessian-nilpotent quartic (382 monomials over $\mathbb{Q}(i)$) and an exact two-point collision, built from Thompson’s dimension-24 cubic-homogeneous form by the de Bondt–van den Essen construction and verified end to end in exact arithmetic.

1 The event, and the discipline applied to it

The counterexample is the polynomial map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$, $u = 1 + xy$,

$$F = (u^3z + y^2u(4 + 3xy), \quad y + 3xu^2z + 3xy^2(4 + 3xy), \quad 2x - 3x^2y - x^3z),$$

with $\det JF = -2$ everywhere and $F(0, 0, -\frac{1}{4}) = F(1, -\frac{3}{2}, \frac{13}{2}) = F(-1, \frac{3}{2}, \frac{13}{2})$ [1]. Our program validated it independently in exact rational arithmetic and computed the full fiber over the repeated value: exactly the three announced points (record EXP-001). Every consequence below is a corollary of this validation plus one or more published implications; for each implication the primary record was located and its direction confirmed before the corollary was asserted (records EXP-016 and EXP-018). Nothing here is original mathematics of ours beyond the verification work and the bookkeeping discipline.

2 The cascade

Corollary 1 (Smale’s 16th problem). *The Jacobian conjecture is false for every $N \geq 3$; the two-variable case remains open.*

Corollary 2 (Mathieu conjecture, 1997). *Mathieu proved that his conjecture for $SU(N)$ implies $JC(N)$ [2, 4]; Duistermaat and van der Kallen proved the abelian case [3]. By contraposition the Mathieu conjecture is **false for $SU(N)$, $N \geq 3$** . It remains open for $SU(2)$ and for the other compact connected groups.*

Corollary 3 (Dixmier conjecture, 1968). *$JC(2n)$ implies $\text{Dixmier}(n)$, and $\text{Dixmier}(n)$ implies $JC(n)$ [5, 6]. Since $JC(4)$ is false, $\text{Dixmier}(2)$ is false: the **full Dixmier conjecture is false**. The original rank-1 case remains open.*

Corollary 4 (Poisson conjecture, 2007). *The Jacobian, Dixmier and Poisson conjectures are equivalent in any characteristic [7]. Hence the **full Poisson conjecture is false**; the minimal failing dimension is open.*

Corollary 5 (Gaussian moments conjecture). *GMC implies JC [8]; hence **GMC is false** at some finite dimension.*

Corollary 6 (Zhao’s vanishing conjecture). *The vanishing conjecture for Hessian-nilpotent polynomials is equivalent to JC [9, 10]; hence it is **false**: there exists a Hessian-nilpotent quartic P with $\Delta^m P^m = 0$ for all m but $\Delta^m P^{m+1} \neq 0$ for infinitely many m . The witness is now explicit: Section 3.*

Corollary 7 (Image conjecture). *The Image conjecture implies the vanishing conjecture [11]; hence it is **false in some dimension**.*

Corollary 8 (Symmetric / gradient Jacobian conjecture). *JC reduces to cubic homogeneous maps with symmetric Jacobian [12]; with the Zhao equivalence this yields a dimension in which a **gradient Keller map is non-invertible**. The map is now explicit, in dimension 48: Section 3.*

3 The explicit witness in dimension 48 (record EXP-041)

As of 2026-07-22 no explicit symmetric, gradient, or Hessian-nilpotent witness had been published (a verified sweep; the July 2026 consequence notes are non-constructive in dimension). The program constructed one, in three verified steps, all exact.

Input. W. Thompson’s explicit Bass–Connell–Wright cubic-homogeneous form of the counterexample [13]: $G = U + H$ on $\mathbb{Q}^{24}$, H cubic homogeneous with 54 monomials, with a published rational collision pair. We re-verified the artifact in-house (homogeneity; the exact collision; nilpotency of JH by an exact sparse chain), with one correction: the nilpotency index is 18, not the claimed 17 ($(JH)^{18} = 0$ while $(JH)^{17}$ retains six terms).

Construction (de Bondt–van den Essen [12]). The potential $f = -i \sum_{j=1}^{24} H_j(x + iy) y_j$ is a homogeneous quartic in 48 variables (382 monomials over $\mathbb{Q}(i)$) with nilpotent Hessian (their characteristic-polynomial identity reduces this to the nilpotency of JH). The triangular conjugation $S^{-1} \circ (\text{id} + \nabla f) \circ S = (x + H(x), y + JH(x)^t y - iH(x))$ was verified identically in all 48 components, so $\tilde{F} = \text{id} + \nabla f$ is a symmetric Keller map ($\det J\tilde{F} = 1$).

The collision. With (u, v) Thompson’s collision pair and $w = i(I + JH(v)^t)^{-1}(u - v)$, the points $p_1 = (u, 0)$ and $p_2 = (v - iw, w)$ are distinct and satisfy $\tilde{F}(p_1) = \tilde{F}(p_2)$ exactly over $\mathbb{Q}(i)$. **Hence the Hessian conjecture fails in dimension 48 with an explicit witness**, and $P_\star = -f$ is an explicit homogeneous Hessian-nilpotent quartic falsifying the vanishing conjecture (the finite certificate of failure is the collision itself; the sanity $\Delta P_\star^2 \neq 0$, forced by Zhao’s own Corollary 3.9 for any failing potential, holds with 9028 terms). The reality obstruction is intrinsic: a nonzero real Hessian-nilpotent polynomial does not exist, so the $\mathbb{Q}(i)$ coefficients are unavoidable. Symmetric positive results (dimension ≤ 4 nonhomogeneous, ≤ 5 homogeneous, de Bondt–van den Essen and van den Essen–Washburn) leave the minimal witness dimension open between those bounds and 48.

4 What remains open

The two-variable Jacobian conjecture; Dixmier at rank 1; Mathieu for $SU(2)$ and the remaining compact groups; the minimal failing dimensions of the Poisson, moments, vanishing and Image statements (for the symmetric/vanishing pair the witness of Section 3 pins the failing dimension at ≤ 48 , with the known positive results leaving the minimum open above 5); and explicit witnesses for the remaining non-constructive rows. Excluded deliberately from this list: the Markus–Yamabe conjecture (refuted independently in 1997, not a corollary of this event) and statements whose implication direction does not propagate the falsity of JC.

5 Method note

The verification program is an open experiment-history repository: every experiment declares a falsifiable hypothesis before running, certifies with exact arithmetic (never floats), closes with a persisted verdict, and keeps refuted attempts in the record. For this document the relevant records are EXP-001 (exact validation, full fiber), EXP-016 and EXP-018 (each chain located in its primary source and its direction confirmed). The general lesson we would offer: when a machine-checkable certificate lands, the fastest reliable way to know what it changes is not commentary but a systematic, source-verified sweep of the implication graph, and that sweep is itself an experiment with a hypothesis, a method and a verdict.

References

- [1] L. Alpöge, announcement of the counterexample, X (@__alpoge__), 2026-07-20; found working with Claude Fable (Anthropic); question posed by Akhil.
- [2] O. Mathieu, Some conjectures about invariant theory and their applications, 1997.
- [3] J. J. Duistermaat, W. van der Kallen, Constant terms in powers of a Laurent polynomial, *Indag. Math.* 9 (1998), 221–231.
- [4] K. Zwart, Mathieu’s approach to the Jacobian Conjecture (review), arXiv:2511.16561.
- [5] A. Belov-Kanel, M. Kontsevich, The Jacobian conjecture is stably equivalent to the Dixmier conjecture, arXiv:math/0512171; *Mosc. Math. J.* 7 (2007).

- [6] Y. Tsuchimoto, Endomorphisms of Weyl algebra and p -curvatures, *Osaka J. Math.* 42 (2005), 435–452.
- [7] P. K. Adjagbo, A. van den Essen, On the equivalence of the Jacobian, Dixmier and Poisson conjectures in any characteristic, arXiv:math/0608009; *Acta Math. Vietnam.* 32 (2007), 209–218.
- [8] H. Derksen, A. van den Essen, W. Zhao, The Gaussian moments conjecture and the Jacobian conjecture, *Israel J. Math.* 219 (2017), 917–928; arXiv:1506.05192.
- [9] W. Zhao, Hessian nilpotent polynomials and the Jacobian conjecture, arXiv:math/0409534; *Trans. Amer. Math. Soc.* 359 (2007).
- [10] W. Zhao, A vanishing conjecture on differential operators with constant coefficients, arXiv:0704.1691.
- [11] A. van den Essen, The amazing Image conjecture, arXiv:1006.5801.
- [12] M. de Bondt, A. van den Essen, A reduction of the Jacobian conjecture to the symmetric case, *Proc. Amer. Math. Soc.* 133 (2005), 2201–2205.
- [13] W. Thompson, Explicit cubic-homogeneous Jacobian counterexample (Bass–Connell–Wright form of the Alpöge map, dimension 24), 2026-07-21; github.com/wtho704/explicit-cubic-homogeneous-jacobian-counterexample; Zenodo record 21466221.